CS342: Computer Networks Lab
(July–November 2025)
Assignment - 01

Submission Deadline: 11:55 pm, 17th August 2025

Wireshark is a free and open-source packet sniffer and network protocol analyzer tool. It helps to capture network packets and understand the structure of different networking protocols. A tutorial for the same can be found here .

1 Instructions:

1. Install Wireshark (download from www.wireshark.org), and learn how to capture packets and filter the required content.
2. A specific application is assigned to groups (refer to Task Allocation Table). Each group needs to perform various activities according to functionalities available in the assigned application and collect the traces for the application using Wireshark. Application-specific activities, if any, are mentioned in the table.
3. You should carry out your experiments across different network conditions including different time(s) of the day and locations (e.g., lab or hostel, etc.).
4. It is advisable to provide only trace-based descriptions while answering the questions. While answering, provide snapshots of the traces in the report and highlight the content as and when required.
5. If something is missing/incorrect in a problem description, clearly mention the assumption with your answer.
6. Submit a soft copy of the report, preferably in PDF format, together with your collected traces in a zip file by the deadline. The ZIP file's name should be the same as your group number.
7. Only one member from a group needs to submit this form by uploading the file.
8. If your trace file size is larger than 2 MB, you are advised to provide the OneDrive/Google Drive/Dropbox link of the traces in your report.
9. The assignment will be evaluated offline/through viva voce during your lab session. where you will need to explain your answers before the evaluator.
10. A plagiarism detection tool will be used and any detection of unfair means will be penalised by awarding NEGATIVE marks (equal to the maximum marks for the assignment).

2 Questions

(a) List out all the protocols used by the allocated application (in the task-allocation table for your group) at different layers (only those which you can figure out from traces). Study and briefly describe their packet formats.

(b) Highlight and explain the observed values for various fields of the protocols. Example: Source or destination IP address and port number, Ethernet address, protocol number, etc.

(c) Explain the sequence of messages exchanged by the application for using the available functionalities in the application. For example: upload, download, play, pause, etc. Check whether there are any handshaking sequences in the application. Briefly explain the handshaking message sequence, if any.

(d) Explain how the particular protocol(s) used by the application is relevant for functioning of the application.

(e) Calculate the following statistics from your traces while performing experiments at different times of the day: Throughput, RTT, Packet size, Number of packets lost, Number of UDP and TCP packets, Number of responses received with respect to one request sent. Use `netstat -s` to observe live protocol statistics, and compare them with your Wireshark traces. Present your findings in a table with observations and time-based variations.

(f) Check whether the whole content is being sent from the same location/source. List out the IP addresses of content providers if multiple sources exist, and explain the reason behind this.

(g) Identify and analyze HTTP GET and POST transactions from your trace. Report the protocols observed, time elapsed between request and response, server and client IP addresses, browser used (User-Agent), and destination port. For the POST request, include the request URL, content type, content length, and data format. Briefly compare GET and POST in terms of data handling and usage. Print both GET and POST messages using Wireshark's "Selected Packet Only" and "Print as displayed" options.

(h) Use nslookup and Wireshark to analyze DNS queries and responses related to IIT Guwahati. Find the IP address of www.iitg.ac.in and the IP address of the DNS server that answered. Identify the name server for the domain iitg.ac.in and how to find its IP address. In Wireshark, locate the DNS query and response for www.iitg.ac.in, and report the transport protocol used, port numbers, and the number of questions and answers. Observe the DNS activity when accessing the IITG website and explain whether DNS caching occurred. Also, perform an nslookup –type=NS iitg.ac.in and summarize the response.

Group No.	**Application**
1–14	Whatsapp
15–28	Google Drive
29–42	Zoom
43–55	Live Sport Streaming
56–68	Youtube - Uploading Video
69–81	MS-Teams
82–94	Youtube - Downloading and Buffering
95–107	Twitch (Live Streaming) or Hotstar Video Streaming

Table 1: Task Allocation Table for Wireshark Experiment

3 Additional Notes

- For video and audio chat-related applications, collect traces with different host locations, (with both the clients within the same network and with one of them outside the LAN) and perform the required analysis.
- Ensure that video uploading and downloading analysis is done with videos that are more than 20 minutes in length.
- To get near-accurate analysis, try to turn off traffic towards unwanted servers, including advertisements and suggestions.
- Do not open any other sites or applications that use the Internet while the packet capture is in progress.
- Use TCPDUMP with necessary filters for actual capture and Wireshark for analysis. Capturing directly with Wireshark causes packets to be lost due to insufficient memory.
- Do not ignore Layer 2 protocols in your analysis.